

PHARMACOVIGILANCE – TASK 3

Causality, Seriousness and Expectedness Assessment

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Course/Internship: Pharmacovigilance Internship

Task: Task 3 – Causality, Seriousness & Expectedness

Product: Velunex

Assessment method: WHO-UMC framework / structured clinical reasoning

1. Assessment Matrix

Case ID	Reported adverse event	Causality assessment	Seriousness & criterion	Severity	Expectedness	Overall rationale
PV-201	Dizziness began 2 hours after first use; resolved same day; no treatment/hospitalization	Possible	Non-serious – no hospitalization, life-threatening event, death, disability or other serious outcome reported	Mild	To be verified against Velunex reference	Strong temporal relationship because dizziness began 2 hours after exposure. However, alternative causes, concomitant products and other clinical information are unavailable. Same-day spontaneous resolution supports a transient event but does not establish causality.

Cas e ID	Reported adverse event	Causality assessment	Seriousness & criterion	Severity	Expectedness	Overall rationale
PV-202	Appendicitis occurred 3 weeks after first use and required hospitalization; medical history incomplete	Unlikely / Unassessable	Serious – hospitalization criterion	Not supplied	To be verified against Velunex reference	Hospitalization makes the event serious. The 3-week interval provides weaker temporal support, and incomplete medical history prevents adequate assessment of alternative causes. Further clinical information is needed.
PV-203	Mild rash began on day 4; improved after stopping; no rechallenge	Probable	Non-serious – no serious outcome or hospitalization reported	Mild	To be verified against Velunex reference	The onset after exposure and improvement after discontinuation provide supportive temporal and dechallenge evidence. However, rechallenge was not performed and alternative causes/concomitant

Case ID	Reported adverse event	Causality assessment	Seriousness & criterion	Severity	Expectedness	Overall rationale
						products are unknown, so causality cannot be considered definite.
PV-204	Loss of consciousness with emergency admission on day 2; concomitant products and test results unknown	Unassessable / Possible	Serious – emergency admission; potentially medically important/life-threatening depending on clinical documentation	Severe	To be verified against Velunex reference	The event occurred after exposure and required emergency admission, but the absence of diagnostic results and concomitant-product information prevents reliable exclusion of alternative causes. Additional clinical information is essential.

2. Detailed Case Assessments

PV-201 – Dizziness

Case information:

Dizziness began two hours after first use of Velunex. It resolved the same day. No treatment or hospitalization was required.

Assessment

- **Causality:** Possible
- **Seriousness:** Non-serious
- **Seriousness criterion:** None identified
- **Severity:** Mild
- **Expectedness:** To be verified against the supplied Velunex reference
- **Dechallenge:** Not specifically reported; event resolved spontaneously
- **Rechallenge:** Not reported
- **Alternative causes:** Unknown

Rationale

The two-hour interval between Velunex exposure and onset of dizziness provides a plausible temporal relationship. However, there is insufficient information about concomitant medicines, medical history, alternative causes, or rechallenge. Therefore, the available information supports possible causality, but not a definite causal relationship.

Missing information / Follow-up

1. What was the dose and route of Velunex?
2. Was the patient taking any concomitant medicines?
3. Did the patient have a history of dizziness, hypotension or vestibular disease?
4. Were vital signs available?
5. Did dizziness recur after subsequent exposure?

PV-202 – Appendicitis

Case information:

Appendicitis occurred three weeks after first use of Velunex and required hospitalization. Medical history is incomplete.

Assessment

- **Causality:** Unlikely / Unassessable
- **Seriousness:** Serious
- **Seriousness criterion:** Hospitalization
- **Severity:** Not supplied

- **Expectedness:** To be verified against the supplied Velunex reference
- **Dechallenge:** Not applicable/not reported
- **Rechallenge:** Not reported
- **Alternative causes:** Cannot be adequately assessed because medical history is incomplete

Rationale

The hospitalization means the case meets a **seriousness criterion**. However, appendicitis occurred three weeks after first exposure, which provides relatively weak temporal evidence for a drug-related reaction. In addition, the medical history is incomplete. Therefore, a causal relationship with Velunex cannot be established from the available information.

Missing information / Follow-up

1. Was appendicitis confirmed clinically or surgically?
2. What was the patient's previous gastrointestinal/medical history?
3. Were any concomitant medicines being taken?
4. Were laboratory/imaging results available?
5. What treatment was given?
6. What was the final outcome?

PV-203 – Mild Rash

Case information:

A mild rash began on day four. It improved after stopping Velunex. No rechallenge was performed.

Assessment

- **Causality:** Probable
- **Seriousness:** Non-serious
- **Seriousness criterion:** None identified
- **Severity:** Mild
- **Expectedness:** To be verified against the supplied Velunex reference
- **Dechallenge:** Positive

- **Rechallenge:** Not performed
- **Alternative causes:** Unknown

Rationale

The rash developed after exposure to Velunex and improved after discontinuation, providing both a plausible temporal relationship and positive dechallenge. These findings support a **probable association**. However, because no rechallenge was performed and information about concomitant products and alternative causes is unavailable, the relationship should not be classified as definite.

Missing information / Follow-up

1. What was the exact type and distribution of the rash?
2. Was there itching, swelling, fever or mucosal involvement?
3. Were any other medicines/products used?
4. Did the rash completely resolve?
5. Was any treatment required?
6. Did the patient have a previous history of allergy?

PV-204 – Loss of Consciousness

Case information:

Loss of consciousness occurred on day two and resulted in emergency admission. Concomitant products and test results are unknown.

Assessment

- **Causality:** Unassessable / Possible
- **Seriousness:** Serious
- **Seriousness criterion:** Emergency hospitalization; potentially medically important/life-threatening depending on documented clinical circumstances
- **Severity:** Severe
- **Expectedness:** To be verified against the supplied Velunex reference
- **Dechallenge:** Unknown
- **Rechallenge:** Unknown
- **Alternative causes:** Cannot be assessed adequately

Rationale

Loss of consciousness requiring emergency admission represents a clinically significant event and meets the seriousness criterion based on hospitalization. The event occurred two days after exposure, which provides some temporal support. However, diagnostic test results and concomitant-product information are unavailable. Syncope/loss of consciousness can have many alternative causes, so the available evidence is insufficient to establish a stronger causal relationship.

Missing information / Follow-up

1. What was the duration of unconsciousness?
2. Were blood glucose, blood pressure and ECG results available?
3. Were neurological investigations performed?
4. Were any concomitant medicines/products being used?
5. Was there a previous history of syncope, seizures or cardiovascular disease?
6. What was the final diagnosis?
7. Did the patient recover completely?
8. Was hospitalization prolonged?

Cas e	Event	Causality	Serious ?	Seriousness criterion	Severit y	Expectedness *
PV- 201	Dizziness	Possible	No	None	Mild	Verify reference
PV- 202	Appendicitis	Unlikely / Unassessabl e	Yes	Hospitalizatio n	Not supplie d	Verify reference
PV- 203	Rash	Probable	No	None	Mild	Verify reference
PV- 204	Loss of consciousnes s	Unassessabl e / Possible	Yes	Emergency admission / medically important	Severe	Verify reference

4. Overall Conclusion

The four cases demonstrate why causality, seriousness, severity and expectedness must be assessed separately.

- **PV-201:** temporally associated dizziness, but insufficient evidence for a definite relationship.
- **PV-202:** serious because hospitalization occurred; causal association is weak/uncertain because of the three-week interval and incomplete medical history.
- **PV-203:** strongest causal evidence among the four cases because of temporal association and positive dechallenge; classified as probable.
- **PV-204:** serious because emergency admission occurred, but causality remains uncertain because important diagnostic and concomitant-product information is missing.

No case should be classified as definitely drug-related based solely on the information supplied.